



AI Product Innovation

Dr. Kornelia Papp

Introduction



Swiss-Hungarian

Lived, studied, and worked
in 9 countries



Mum



PhD in Cognitive Semantics

15+ years in industry

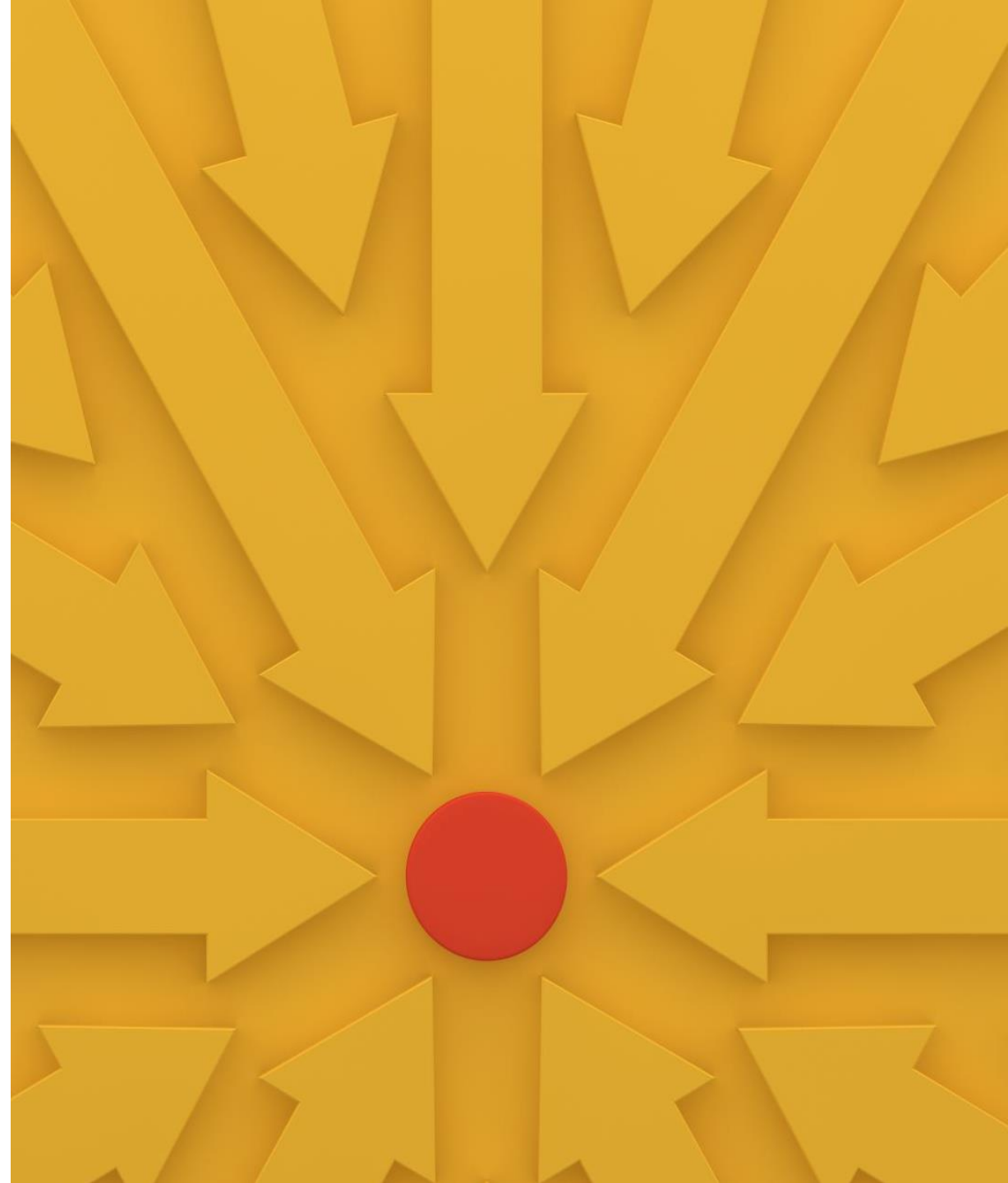
Founder of NLP Zurich
Applied AI Lecturer



Invitation to connect



Unlock the
potential of AI and
use it as a catalyst
to **build better
products.**



What is a product?

The word "product" is a broad umbrella. It essentially describes anything that is **intentionally designed and packaged to provide value**.

- **Physical Goods:** The obvious ones (phones, coffee, sneakers).
- **Digital Services:** Apps, websites, and even the AI models they are learning about.
- **Experiences:** A music festival, a flight from London to New York, or a university course.
- **Processes:** A specific way of onboarding new employees or a unique method for cleaning data.
- **The "Self":** In a job interview, your skills, time, and personality are the "product."

What is a product?

Your "Product"

How AI Makes it Better

Engineer	The Codebase / API	AI handles the syntax and unit tests, allowing you to focus on system architecture and security.
Designer	The UI / Brand Identity	AI generates endless variations and mock-ups, allowing you to focus on human-centric empathy and visual story.
Manager	Team Efficiency / Roadmap	AI synthesizes meeting notes and data trends, allowing you to focus on removing blockers and high-level strategy.
Actuary	The Risk Model	AI runs complex simulations at scale, allowing you to focus on human judgment and ethical implications.

A product isn't just something you put in a box or download from a store. A product is **value made tangible**. If it solves a problem and can be improved with AI, it's a product.

Learning structure

01	Topic introduction and context setting
02	Iterative learning and refinement
03	Quiz
04	Exercises

Location: HG G 19.1 (G Floor in the ETH Main Building).

Date: 29 May, 30 May

Time: 9:15-17:00 (with breaks)

How we work together

- self-driven
- curious
- participate actively
- support peers
- share ideas and questions openly
- behave respectfully and constructively



Prerequisites

Students should be able to:

- write and modify simple Python code
- understand basic traditional machine learning concepts
- apply known concepts to simple practical problems

Set up:

- Access to embeddings (e.g., text-embedding-3-small)
- Hugging Face account
- Bloom account and Bloom: <https://bloom.diy/>
- Claude Code (I will use it in app and in VS Code): <https://code.claude.com/docs/en/quickstart>



Technical setup

You can use your favourite IDE. In the class, I will be using VS Code with Claude. If you are not an experienced coder, I recommend having the same set up.

Instructions:

Add Claude Code Extension in VS Code.

At least Claude Code PRO Subscription (ca. 20 CHF) will be needed.

Exclusive: Meta targets May 20 for first wave of layoffs; additional cuts later in 2026

By Katie Paul and Jeff Horwitz

April 17, 2026 9:42 PM GMT+2 · Updated April 20, 2026



Cars drive past a sign of Meta, the new name for the company formerly known as Facebook, at its headquarters in Menlo Park, California, U.S. October 28, 2021. REUTERS/Carlos Barria/File Photo [Purchase Licensing Rights](#)

Summary Companies

- First wave to cut about 10% of Meta's workforce, sources tell Reuters
- Further layoffs planned for later in 2026, details not finalized
- Layoffs linked to [AI-driven efficiency push](#), mirroring broader tech-sector trends

NEW YORK/SAN FRANCISCO, April 17 (Reuters) - Meta ([META.O](#)) intends to conduct a first wave of sweeping layoffs planned for this year on May 20, with more coming later, three sources familiar with the plans told Reuters.

The Facebook and Instagram owner will lay off about 10% of its global workforce, or close to 8,000 employees, in that initial round, one of the sources said.

Gadget Review...

The AI Termination Ban: Why Chinese Courts Just Made It Illegal to Replace Workers with Robots

AI Landes

Thu, April 30, 2026 at 8:40 PM GMT+2 · 2 min read



Image: Flickr - Gary Lerude

Your favorite gadgets just got more expensive to make—and that might be a good thing. Chinese courts delivered a surprising blow to tech companies betting on AI-powered cost-cutting, ruling that [artificial intelligence adoption doesn't justify terminating employees under the country's Labor Contract Law](#).

The evolution of your role

Human Moat

Define your role, identify the tasks AI will automate, and then describe your 'moat'. The uniquely human skills (empathy, ethics, physical dexterity, complex intuition) that keep your role relevant.

AI Co-Worker

What part of your job would you happily give to an AI assistant tomorrow, and what part would you never let it touch, even if the tech was perfect?



Example: Junior Graphic Designer

Category	Junior Graphic Designer
Core Mission	Translating abstract ideas into compelling visual stories that solve specific problems.
Delegate to AI	Asset cleanup, colour/font iterations, production mock-ups, and stock photo customization.
Keep	Conceptual "Why," interpreting vague client feedback, strategic originality, and visual ethics.
The Shift	From being a pixel-pusher to being a visual strategist .

Example: Head of AI Product Development

Category	Head of AI Product Development
Core Mission	Bridging the gap between technical capability and user value to build meaningful products.
Delegate to AI	Data synthesis, drafting PRDs, code scaffolding, and routine stakeholder updates.
Keep	Navigating ambiguity, stakeholder diplomacy, ethical guardrails, and defining the "North Star."
The Shift	From managing execution to managing vision and ethics .